

Machine Learning Algorithm Based Prediction of Players' Scores in Tennis Matches

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Abstract—During the men's singles final of the 2023 Wimbledon Open, the match situation fluctuated dramatically, and "momentum" was cited as the cause of these fluctuations. In order to specifically describe and analyze the impact of "momentum" on the match situation, first, we created a set of metrics and applied logistic regression for significance analysis to clearly demonstrate that the system has a significant impact on players' real-time scoring situations. In addition, we used several machine learning algorithms and cross-validation methods based on this system to test the algorithm performance. As a result, we employ LGBM models to predict player "momentum" while providing visualizations of the predicted results. Preliminary Pearson's correlation tests confirm that there is a significant correlation between "momentum" and a player's actual score. In addition, a simple linear regression model was constructed with "momentum" as the independent variable and the player's actual score as the dependent variable. Therefore, it can be concluded that the view expressed by tennis coaches that "a player's swing and success in a match is random" is biased.

Keywords—Machine Learning, Tennis Match, LGBM, Pearson Correlation Coefficient

I. INTRODUCTION

In the 2023 Wimbledon Men's Singles Final, Spanish star Carlos Alcaraz faced legendary veteran Novak Djokovic in a major change of scene. In the first four sets, they were tied 2-2. In the fifth set, Alcaraz won 6-4 to win the title and end Djokovic's winning streak at Wimbledon since 2013. During the course of the match, the performances of both players fluctuated, showing different trends at different times. We analyzed this so that athletes can feel and utilize it in their sport. Understanding how momentum affects an athlete's performance and game trends can greatly facilitate our prediction of the game [1].

In order to better determine which player is performing better at a given time and the probability that they will win, a game process capture and evaluation model was built to record the scores of each game in detail. This model can be applied to describe the game process visually. Finally, a "momentum" evaluation model was developed to verify the significant effect of momentum on players' scores in matches using the model and indicators developed, and to dispel the doubts of tennis coaches about the role of "momentum" in matches [2].

II. EVALUATION MODEL OF THE GAME PROCESS

We searched for various factors and indicators that may have an impact on the player's score by investigating relevant information. After screening and induction, we summarized and ultimately constructed a complete and comprehensive indicator system, which mainly includes the fatigue level of

players, personal technical ability, and real-time mentality of the game. We applied this system to more comprehensively measure the real-time scoring ability of players.

Firstly, we conducted a significance test on the correlation between the constructed indicator system and labels using the LOGISTIC method. Analyzing these indicators and the labels of whether players have scored in the tennis game dataset provides us with sufficient data support, which helps to analyze and compare the impact of various factors on player scores, and also provides a basis for feature selection in subsequent modeling.

Next, we utilized various machine learning algorithms, including LGBM, support vector machines, perceptron networks, logistic regression, and XGBOOST. Using 5-fold cross validation, we comprehensively evaluated the strengths and weaknesses of these models, and compared them using metrics such as precision, fl, accuracy, recall, AUC, and ROC curves. This approach not only ensures the robustness and generalization ability of the model, but also provides strong support for the selection of the final modeling algorithm.

In the end, we chose the best LGAM algorithm for performance evaluation for modeling. This algorithm not only dynamically evaluates the real-time performance of players, but also locks in the "momentum" changes during the competition. By using this model, we can more accurately predict the performance of players in the competition and judge the results of the competition more accurately. Through the above analysis, we have made significant progress in analyzing and evaluating the real-time tactical score performance of players, providing a foundation and direction for further analysis and prediction.

TABLE I. INDEX SYSTEM OF FACTORS INFLUENCING PLAYER TACTICAL SCORES

Indicator	Classification
The number of games won in a set	individual ability
leading score in this game	individual ability
Whether to serve or not	serving real-time situation
Did you score in the previous round	individual ability/implementation situation
Did you score in the previous round	individual ability
The leading progress in sets in this game	individual ability/state
Does this game score a counterattack (no touch)	individual ability/state
Has there been a double error in this bureau	state
Has there been any non mandatory double error in this bureau	state
The proportion of internet usage and scores obtained during internet usage	Exhaustion state/state
The total running distance of this competition	Exhaustion state
Total mileage of the last three rounds	Exhaustion state
Mileage of the previous round's running chart	Exhaustion state
Real time serving speed	state
Is it an interaction term between the server and the real-time speed of the server	state

A. Construction of the indicator system

We firstly established a preliminary judgment index system of possible factors affecting players' scoring by reviewing relevant information and understanding the related matters of tennis matches, as shown in Table I. The system mainly considers four specific categories of relevant indicators, namely, individual ability, real-time situation, fatigue state, and psychological state.

B. Statistical logistic test based on indicator system

According to this indicator system, we can predict the real-time scoring ability of any player in any round of any competition. To measure the impact of different indicators on the scoring ability of each round, we adopt statistical regression to verify the impact of each indicator on the scoring situation. Since there are only two scoring scenarios: scoring or losing points, we naturally think of applying a binary logistic regression model for binary variables and using SPSS software to output the results.

The results are shown in Table II.

TABLE II. CLASSIFICATION PERFORMANCE OF LOGISTIC REGRESSION

classification table					
	actual value		predicted value		
			label		Correct percentage
	0	1			
Step 1	label	0	271	517	34.4
		1	184	1058	85.2
	Average percentage				65.5
a. boundary value .500					

According to the statistical results, the overall accuracy of the prediction results obtained by the logistic regression model is 65.5%. Specifically, the accuracy of the model's prediction without scoring is 34.4%, while the accuracy of the predicted score is 85.2%. From this, it can be seen that the current model is more likely to judge the sample towards a score of label=1. However, the purpose of establishing the current model is to conduct significance analysis on the constructed indicators, rather than obtaining the final model. The test results are shown in the Table III.

TABLE III. VARIABLE TABLE

	B	Standard error	Wald	DOF	significance	Exp(B)
x1	0.016	0.176	0.009	1	0.926	1.017
x2	-1.141	0.363	9.881	1	0.002	0.319
x3	0.689	1.286	0.287	1	0.592	1.992
x4	0.170	0.166	1.048	1	0.306	1.186
x5	0.032	0.203	0.024	1	0.876	1.032
x6	0.689	0.127	29.400	1	0.000	1.991
x7	0.456	0.128	12.697	1	0.000	1.577
x9	0.472	0.153	9.491	1	0.002	1.603
x10	-0.535	0.109	24.287	1	0.000	0.586
x11	0.914	0.124	54.371	1	0.000	2.493
x12	0.073	0.140	0.273	1	0.601	1.076
x13	0.120	0.288	0.174	1	0.676	1.128
x14	-0.164	0.482	0.116	1	0.734	0.849
x15	-1.445	0.657	4.842	1	0.028	0.236
x16	0.511	0.626	0.668	1	0.414	1.667
const	-0.876	1.693	0.268	1	0.605	0.416

When the significance calculation value of the independent variable is less than 0.05, we have reason to believe that the independent variable can significantly affect the dependent variable. Therefore, the x2, x6, x7, x8, x9, x10,

and x14 independent variables in the table can significantly affect the player's real-time scoring situation. Therefore, referring to the previous classification of individual variables, we can obtain the player's personal ability, fatigue state, and psychological state. Both can significantly affect the player's real-time scoring ability.

C. Machine learning model validation based on indicator system

LGBM is similar to GBDT and XGBoost in principle, which use the negative gradient of the loss function as the residual approximation of the current decision tree to fit a new decision tree. LGBM adopts the hisLogram algorithm, and the basic idea of the Histogram algorithm is to discretize the continuous floating-point eigenvalues into k integers, and at the same time, construct a histogram with width k. When traversing the data, according to the discretized values as indexes, the histogram accumulates the required statistics, and after traversing the data once, the histogram accumulates the required statistics. When traversing the data, the discretized values are used as indexes to accumulate statistics in the Histogram, when traversing the data once, the Histogram accumulates the required statistics, and then traverses the data to find the optimal segmentation point based on the discretized values of the Histogram [5].

Suppose there are n instances $x_1, \dots, x_n$ in the training set. At each gradient iteration, the model analyzes the data to study the direction of the negative gradient development of the loss function of the variables can be expressed as $g_1, \dots, g_n$. The decision tree divides the data into different nodes through an optimal splitting point (the point of maximum information system gain) [3]. GBDT measures the information gain by the variance after segmentation. For example, O denotes a training set of fixed nodes, d denotes the segmentation of feature j, defined as.

$$V_{j|O}(d) = \frac{1}{n_O} \left(\frac{(\sum_{x_i \in O, x_{ij} \leq d} g_i)^2}{n_{l|O}^j(d)} + \frac{(\sum_{x_i \in O, x_{ij} > d} g_i)^2}{n_{r|O}^j(d)} \right) \quad (1)$$

$$\text{Where, } n_O = \sum [x_i \in O] \text{ , } n_{l|O}^j = \sum I_i [x_i \in O_{:x_i} \leq d] \text{ , } n_{r|O}^j = \sum I_i [x_i \in O_{:x_i} > d]$$

Iterate each split point of each feature, find $d_j^* = \arg\max_d V_j(d)$, and calculate to get the maximum information system gain $V_j(d_j^*)$, and then, the data can be divided into left and right child nodes according to the different features j^* of the split point d_j^* will be carried out [4].

In order to establish an accurate machine learning model for the indicator system, we selected the ensemble tree models LGBM and XGBOOST, which currently have good accuracy performance and training effects, and are not easy to overfit. We compared them with classic machine learning algorithms such as support vector machines, perceptron networks, and logistic regression. Then, the cross validation method is applied to validate the model, and the accuracy, precision, recall, and fl based on the confusion matrix, as well as the AUC and ROC curves, are used to evaluate the results. The results based on the confusion matrix are shown in Table IV, and the training and testing sets based on the AUC and ROC curves are shown in Figure 1 and Figure 2, respectively. Table IV shows Cross-validation algorithm results.

TABLE IV. CROSS-VALIDATION ALGORITHM RESULTS

	acc	recall	precision	f1	auc
LGBM	0.69	0.69	0.7	0.69	0.77
XGB	0.67	0.68	0.68	0.68	0.75
SVC	0.67	0.65	0.7	0.67	0.75
MLP	0.69	0.66	0.71	0.68	0.76
LR	0.67	0.69	0.67	0.68	0.72

From the data in the table, it can be seen that the best performing model is LGBM, with accuracy, precision, recall, f1, and auc of 0.69, 0.69, 0.7, 0.69, and 0.77, respectively. Next is the perceptron neural network, with values of 0.69, 0.66, 0.71, 0.68, and 0.76. The difference between different indicators is very small, indicating that there is no significant preference for sample discrimination in this model, which further proves that LGBM performs the best.

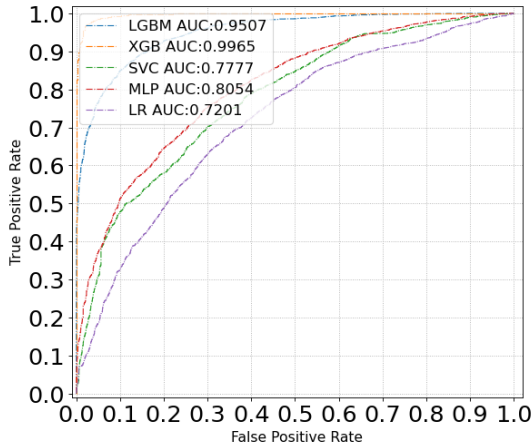


Fig. 1. ROC curve of training set

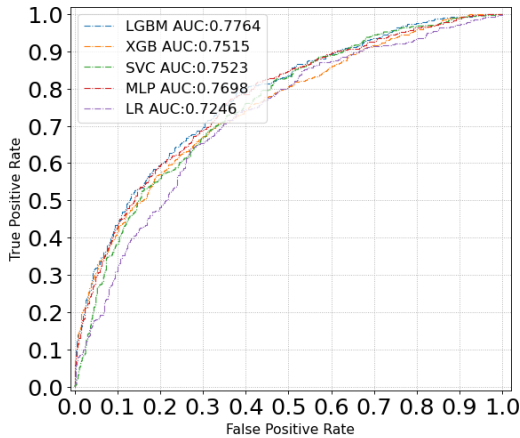


Fig. 2. ROC curve of test set

We tested the changes in precision and recall indicators under different thresholds using ROC curves, and based on the ROC curves, we can also determine that LGBM is the model with better performance. Based on this conclusion, we use the LGBM model with better regression performance to visualize the real-time scoring performance of Carlos Alcaraz against Novak Djokovic in the 2023 match given in this question. The selected recorded player is Carlos Alcaraz, and the results are shown in Figure 3.

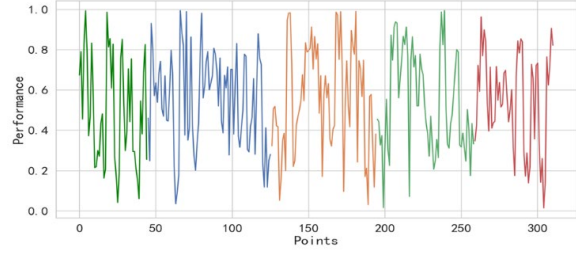


Fig. 3. The results of Carlos Alcaraz

From the results, it can be seen that the model can only accurately predict the actual score of the 0.7 part of the players. According to our analysis, due to the various types of actual factors that determine the scores of players and the possibility of unexpected events, if these issues cannot be taken into account more, the prediction results of the player's score model will inevitably have more serious "noise". However, at the same time, we found that the server and scorer of this classic game were exactly consistent with the player's actual scoring situation, while Carlos Alcaraz had a higher momentum when winning and a smaller momentum when losing. It can be concluded that the model proposed in this article is still effective.

III. MOMENTUM ASSESSMENT MODEL

The method we use is Pearson correlation test, which first applies the best machine learning model to the test set to obtain the player's scoring probability. By conducting a Pearson correlation test between the obtained probability and the actual score, this study explores the correlation between the scoring probability and the actual score, in order to preliminarily evaluate whether "momentum" truly plays a role in the competition process.

To further investigate, we used univariate linear regression to model the problem, with "momentum" as the independent variable of the model and player actual score as the dependent variable. Through this momentum evaluation model, not only can we explore whether and to what extent momentum plays a role in a player's actual score, but we can also explore the interpretable part behind the impact of momentum on actual score. This in-depth and thorough problem analysis process helps to explore the exact impact mechanism of "momentum" on player scoring, thereby more comprehensively visualizing the dynamic characteristics of player performance during the game.

Under this research framework, the expected result is that "momentum" is not only a simple random phenomenon, but in fact plays a very important and significant role in player scoring. Through the above research process, we have gained a deeper and more detailed understanding of the practical significance and role of "momentum" in the game process. This can provide constructive and specific game suggestions and guidance for players during the game, enabling them to achieve their best performance.

A. Pearson correlation test

Pearson correlation coefficient is a statistical measure used to measure the linear relationship between two variables. Its value range is between -1 and 1, where:

-1 represents complete negative correlation: that is, when one variable increases, the other variable decreases accordingly.

1 represents complete positive correlation: that is, when one variable increases, the other variable increases accordingly.

0 indicates no linear relationship: that is, there is no linear relationship between two variables.

The calculation formula for Pearson correlation coefficient (usually represented by the symbol r) is as follows:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}} \quad (2)$$

Among them: X_i and Y_i are corresponding data points. And $\bar{X}$ and $\bar{Y}$ are the mean values of the corresponding variables. The numerator represents the sum of the product of the difference between each data point of two variables. The denominator represents the square root of the product of the sum of squared differences between each data point of two variables.

The closer the calculated r value is to 1 or -1, the stronger the linear relationship between the two variables. The closer the value is to 1, the stronger the positive correlation; The closer the value is to -1, the stronger the negative correlation; If the value is close to 0, it indicates a lack of linear relationship between the two variables.

In the modeling process, it is expected that "momentum" is not only a simple random phenomenon, but also plays a very important and obvious role in player scores. Therefore, it is necessary to first verify that the correlation between "momentum" and player actual scores is significant. The results are shown in the Table V.

TABLE V. CORRELATION BETWEEN MOMENTUM AND PLAYER POINT SCORES

"Correlation coefficient."	P-value
0.482	0.000

B. Univariate linear regression results

The impact of "momentum" on players' actual scores was analyzed using a univariate linear regression analysis method, and the results are shown in the Table VI.

TABLE VI. RESULTS OF UNIVARIATE LINEAR REGRESSION

	coef	std err	t	P> t	[0.025	0.975]
const	0.0320	0.027	1.176	0.240	-0.021	0.085
x1	0.9231	0.048	19.406	0.000	0.830	1.016
Omnibus:	459.412	Durbin-Watson:	2.088			
Prob(Omnibus)	0.000	Jarque-Bera(JB)	62.513			
Skew:	0.034	Prob(JB):	2.66e-14			
Kurtosis:	1.904	Cond.No	4.88			

From the graph, it can be seen that the P-value obtained is also less than 0.05, and its weight is 0.9231. The correlation between "momentum" and the actual score of players is significant. The influence of "momentum" on the actual scoring situation of players is very significant, which can explain the 0.233 obtained in the results.

IV. CONCLUSION

After investigation and analysis, we determined that the score mentioned in the question is not only related to whether the player is the serving side or not, but also the real-time fatigue level of the player, individual technical and tactical ability, and the change of the player's mentality during the

match are important reference factors that affect the player's score. Therefore, we established an index system based on the above reference factors, introduced and comprehensively analyzed the influence and relative importance of each index on whether the player scored or not, and based on this, we made a comprehensive evaluation of the player's real-time score. First, we calculate the above indicators and labels based on the tennis match dataset, and then test the significance of the indicator system and labels by statistical LOGISTIC analysis. Next, we evaluate the performance of the model using various machine learning algorithms such as LGBM, Support Vector Machines, Perceptron Networks, Logistic Regression, and XGBOOST using 5-fold cross-validation and precision, fl, accuracy, recall, auc, and roc curves. Finally, based on the evaluated performance, we select the best performing LGAM algorithm for modeling and dynamically evaluate the real-time performance of the participants, i.e., "momentum".

The real-time performance evaluation index system we constructed for tennis players also has certain applicability and universality in other sports competitions. Although different sports have their unique rules and characteristics, certain key factors and performance indicators have commonality among different sports. Considering the above factors, although it is necessary to make moderate adjustments according to the rules and characteristics of specific sports, our indicator system is based on general sports principles and can provide a useful reference framework for other sports competitions. Through flexible application and adjustment, the indicator system can provide strong support for real-time performance evaluation of athletes in other sports.

In order to analyze the role of "momentum" in the game and to determine whether the fluctuation of players' performance and the phenomenon of success in the game are non-random in nature, we first used the best model established to obtain results in the test set and to test the Pearson's correlation between the obtained score probabilities and the actual scoring situation. Then, we constructed a univariate linear regression model with "momentum" as the independent variable and the dependent variable as the real scores, expressed the abstract concept of "momentum" with objective and practical parameters, and proved that "momentum" has a significant effect on players' scores.

For coaches, understanding and utilizing the Momentum indicator system can provide targeted tactical and training advice to maximize player performance.

REFERENCES

- [1] Zhang Rong. Tennis match player prediction and analysis [D]. Yunnan university, 2023. the DOI: 10.27456 /, dc nki. Gyndu. 2022.000807.
- [2] Huang Zhiying. Research on prediction model of tennis match results based on BP neural network [D]. Fujian normal university, 2022. doi: 10.27019 /, dc nki. gfjsu. 2022.000427.
- [3] Y.H. Liu,X.F. Qu. Machine learning LGBM regression algorithm for predicting concrete strength[J]. Sichuan Construction,2021,41(6):178-180.
- [4] Li Hui. Soft measurement method of cement fineness based on BP neural network[J]. China Cement,2016(7):107-109.
- [5] TANG Huasong,YAO Yaowen. Exploration of decision tree algorithm in data mining[J]. Computer Application Research,2001,18(8):18-19.